

Proposal

Team Project Proposal

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Project Title:

Studying the Effect of the Weather on NYC Transportation Frequency (Citi Bike, Yellow Taxi)

Project Description:

This project will analyze transportation usage patterns in New York City by examining Citi Bike trip data, yellow taxi trip records, and weather data from the year 2019. The transportation datasets contain information such as trip start time, trip duration, pickup and drop-off locations, and ride frequency, which makes it possible to compare how different transportation methods are used across time and across different parts of the city. The weather data we are using is specific to day and the neighborhood in which temperature was recorded. By combining these three sources, our analysis will focus on how residents and visitors choose between bike sharing and taxi services depending on weather, location, time of day, and any other correlating features.

Our research question for this project is: how do different weather conditions (temperature, precipitation, and extreme weather events) affect the relative frequency of Citi Bike usage compared to yellow taxi trips across New York City in 2019? This question will be explored by comparing trip volumes over time, examining differences between Manhattan and the outer boroughs, and identifying patterns that may suggest whether Citi Bike is used as an alternative to traditional taxi transportation.

Who will benefit from your analysis?

Since people generally don't own cars in the city, they rely on alternative modes of transportation instead. Through the results of our analysis, city urban planners can potentially optimize citi-bike station placement, taxi stand locations or identify underserved neighborhoods to better meet supply and demand.

Additionally, policymakers can receive valuable insights to understand how to allocate resources more efficiently during bad weather.

In conclusion, the public would benefit because transportation would be more convenient and there would be less overcrowding of vehicles including bikes on the road.

What insight will you derive from the data?

Our primary insight will be the determining the contributing factors that lead to shift from taxi to Citi Bikes. We will analyze cost, distance, weather, time of day, and location to assess the specific point at which a bicycle becomes a more optimal choice than an automobile in terms of time and expense. Furthermore, we will identify locations where additional Citi Bike docks could facilitate easier transportation for civilians by evaluating bike usage over various boroughs. Finally, we will determine the effect of bikes on congestion, comparing Citi Bike usage to traffic to see if increased volume correlates with improved traffic in New York city.

Describe how you will check the goodness of the analytic. i.e., why do you believe that the results are accurate and can be trusted? Include references to research papers. Share at least one research paper per team member. (see <https://scholar.google.com/>)

Our sources come from NYC Open Data, which comes directly from official sources. However, to further ensure that results are accurate and can be trusted, we will extensively clean data by removing outliers, such as Citi Bike trips under 30 seconds or over 12 hours as these often represent users testing bikes or data entry errors, rather than meaningful trips.

To make our results more accurate, we will focus on 1 year's worth of data and then compare the months in that data. This will be more efficient in that we are reducing the data needed to analyze as well as citi bike's data is separated based on months rather than years. To find consistent patterns, we will divide up the month into 2 segments (early and late segments) and do a comparison between them. Additionally, if there are any MV's (missing values) then we will replace them with the average of the rows to provide more accurate readings. To find the most recent transportation trends, we will find the most recent year out of all 3 datasets that have sufficient data to analyze.

- [UNDERSTANDING USAGE OF PUBLIC BIKE SHARING SYSTEM : CITI BIKE AS AN EXAMPLE](#) - Matthew
- [Predicting Bike Usage for New York City's Bike Sharing System](#) - Adam
- [Modeling and Analyzing the Spatiotemporal Travel Patterns of Bike Sharing: A Case Study of Citi Bike in New York](#) - Andrew
- <https://link.springer.com/article/10.1007/s12652-018-0820-5> - Andrew

Data Sources:

*Include the source, link and schema of the datasource. *Please do not use Kaggle* Please try to find large data sources.*

Schema look like: Field Name, Data type, Brief description

- [Citi Bike System Data \(NYC\)](#) – Matthew

Ride ID, String, Unique identifier for each individual trip
Rideable type, String, Type of bike used (Classic or Electric)
Started at, Timestamp, The date and time the trip began
Ended at, Timestamp, The date and time the trip ended
Start station name, String, Public name of the station where the trip started
Start station ID, String, Unique identifier for the starting station
End station name, String, Public name of the station where the trip ended
End station ID, String, Unique identifier for the destination station
Start latitude, Float, The GPS latitude of the starting station
Start longitude, Float, The GPS longitude of the starting station
End latitude, Float, The GPS latitude of the destination station
End longitude, Float, The GPS longitude of the destination station

Member or casual ride, String, Indicates if the rider is an annual member or a casual user

- [2019 Yellow Taxi Data \(nyc.gov TLC - Taxi and Limousine Commission\)](#) — Adam
 - [Dictionary](#)

VendorID, Number, A code indicating the TPEP provider that provided the record

tpep_pickup_datetime, Floating Timestamp, The date and time when the meter was engaged

tpep_dropoff_datetime, Floating Timestamp, The date and time when the meter was disengaged

passenger_count, Number, The number of passengers in the vehicle

trip_distance, Number, The elapsed trip distance in miles reported by the taximeter

RatecodeID, Number, The final rate code in effect at the end of the trip

store_and_fwd_flag, Text, Indicates if the trip record was held in vehicle memory before sending to the vendor

PULocationID, Number, TLC Taxi Zone in which the taximeter was engaged

DOLocationID, Number, TLC Taxi Zone in which the taximeter was disengaged

payment_type, Number, A numeric code signifying how the passenger paid for the trip

fare_amount, Number, The time-and-distance fare calculated by the meter

extra, Number, Miscellaneous extras and surcharges

mta_tax, Number, Tax automatically triggered based on the metered rate in use

tip_amount, Number, Tip amount automatically populated for credit card tips

tolls_amount, Number, Total amount of all tolls paid in trip

improvement_surcharge, Number, Improvement surcharge assessed trips at the flag drop

total_amount, Number, The total amount charged to passengers (excludes cash tips)

congestion_surcharge, Number, Total amount collected in trip for NYS congestion surcharge

airport_fee, Number, Fee for pick up only at LaGuardia and John F. Kennedy Airports

- [Weather \(NYC OpenData\)](#) — Andrew

Sensor.ID,unique identifier of sensor,sensor_id,Text

AirTemp,average hourly air temperature Fahrenheit,airtemp,Number

Day,Date,day,Floating Timestamp

Hour,Hour of day,hour,Number

Latitude,Latitude of sensor,latitude,Number

Longitude,Longitude of sensor,longitude,Number

Year,Year,year,Text

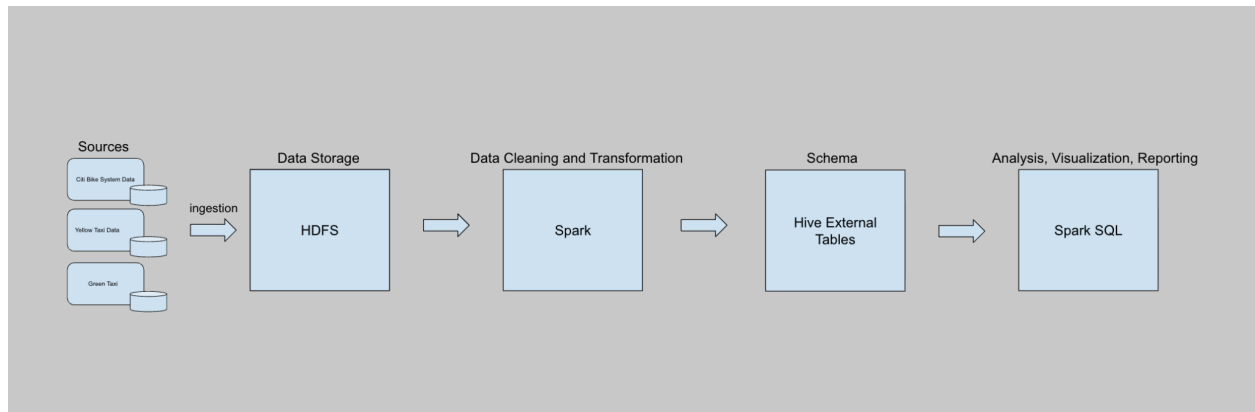
Install.Type,Type of mounting (street tree/street light),install_type,Text

Borough,borough,borough,Text

ntacode,Neighborhood Tabulation Areas (NTA) code,ntacode,Text

Draw initial design diagram(s) (ie. pipeline), using PowerPoint, Keynote, Google Drawing, etc. to describe your project work flow.

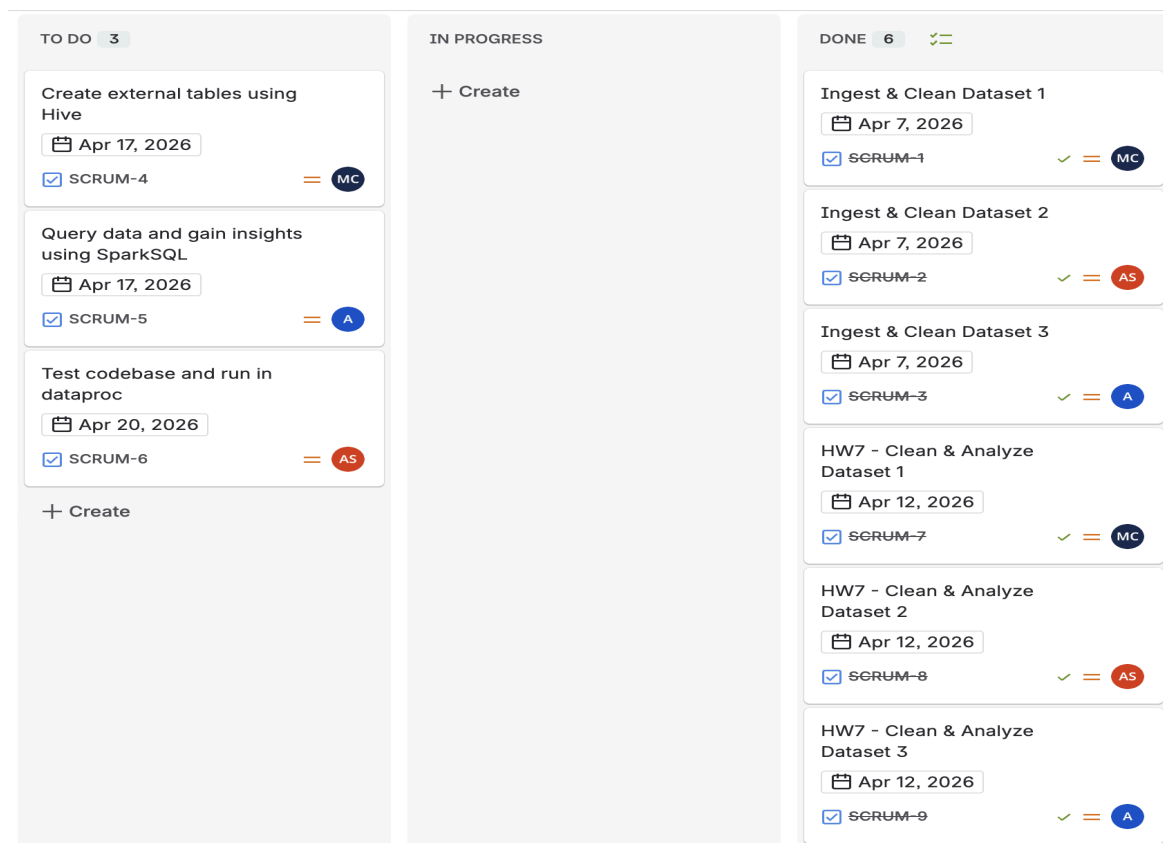
Include the software architecture (Big Data tools you'll use), the data flows, and anything else you think is important to show. Your project will run on NYU's HPC Dataproc.



BDA - Data Pipeline

Task List: Jira

Assign team members to tasks, and assign a due date to each task. This can be just a simple story with a titled task, assigned a team member name, and target completion date. Provide screenshots of your team's sprint, even if you are one person. Create not just immediate tasks but ones you need to complete for this project (profiling data, cleaning data,..).



Jira Screenshot for HW7

Questions

Questions

Deadlines:

- Apr. 20 – final code submission
- Apr. 22 – final slides

Other dataset ideas...

- Subway locations
- Pedestrian accident
- Car accident: more than 1 car
- Bad Road conditions
- Weather
- Accidents: how it affects traffic in the city?

nyc.gov monthly taxi [data](#) yellow/green (Green taxi is limited)

[Pothole dataset](#)

[Automated traffic counts dataset](#)

[NYPD Complaint Data Historic](#)

[School Quality Reports Data](#)

SF & NYC 311 cases vs economic data

- Are more crimes committed in periods of economic struggle or success?
- [SF 311 cases](#)
- [NYC 311](#)
- [NASDAQ Daily index](#)

Hive hdfs commands

```
# hdfs dfs -ls /user/hive/warehouse/ | grep mc9967_nyu_edu
drwx----- - mc9967_nyu_edu mc9967_nyu_edu          0 2026-01-22 13:33
/user/hive/warehouse/mc9967_nyu_edu.db
```

```
# hdfs dfs -ls /user/mc9967_nyu_edu/ | grep citibike_cleaned
drwxr-xr-x - mc9967_nyu_edu mc9967_nyu_edu          0 2026-04-04 01:16
/user/mc9967_nyu_edu/citibike_cleaned
```

You should be able to run the command to grant permissions. Please see:

<https://services.rt.nyu.edu/docs/cloud/dataproc/computation/#using-hive>
and the command example:

```
hdfs dfs -setfacl -R -m user:<OTHER_PERSON_NETID>_nyu_edu:r-x
/user/hive/warehouse/NetID_nyu_edu.db
```